**SECTION 1 (Maximum Marks: 24)**

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +3 **ONLY** if the correct numerical value is entered;
*Zero Marks* : 0 In all other cases.

Q.1 Considering only the principal values of the inverse trigonometric functions, the value of

$$\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\,\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$$

is _____ .

Q.2 Let $\alpha$ be a positive real number. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:(\alpha,\infty)\to\mathbb{R}$ be the functions defined by

$$f(x)=\sin\left(\frac{\pi x}{12}\right) \qquad \text{and} \qquad g(x)=\frac{2\log_e\left(\sqrt{x}-\sqrt{\alpha}\right)}{\log_e\left(e^{\sqrt{x}}-e^{\sqrt{\alpha}}\right)}.$$

Then the value of $\lim_{x\to\alpha^+} f(g(x))$ is _____.

Q.3 In a study about a pandemic, data of 900 persons was collected. It was found that

190 persons had symptom of fever,
220 persons had symptom of cough,
220 persons had symptom of breathing problem,
330 persons had symptom of fever or cough or both,
350 persons had symptom of cough or breathing problem or both,
340 persons had symptom of fever or breathing problem or both,
30 persons had all three symptoms (fever, cough and breathing problem).

If a person is chosen randomly from these 900 persons, then the probability that the person has at most one symptom is _____.

Q.4 Let $z$ be a complex number with non-zero imaginary part. If

$$\frac{2+3z+4z^2}{2-3z+4z^2}$$

is a real number, then the value of $|z|^2$ is _____.

Q.5 Let $\bar{z}$ denote the complex conjugate of a complex number $z$ and let $i=\sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation

$$\bar{z}-z^2=i(\bar{z}+z^2)$$

is _____.

Q.6 Let $l_1, l_2, \ldots, l_{100}$ be consecutive terms of an arithmetic progression with common difference $d_1$, and let $w_1, w_2, \ldots, w_{100}$ be consecutive terms of another arithmetic progression with common difference $d_2$, where $d_1 d_2 = 10$. For each $i = 1, 2, \ldots, 100$, let $R_i$ be a rectangle with length $l_i$, width $w_i$ and area $A_i$. If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is _____.

Q.7 The number of 4-digit integers in the closed interval $[2022, 4482]$ formed by using the digits $0, 2, 3, 4, 6, 7$ is _____.

Q.8 Let $ABC$ be the triangle with $AB = 1$, $AC = 3$ and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides $AB$, $AC$ and also touches internally the circumcircle of the triangle $ABC$, then the value of $r$ is _____.

**SECTION 2 (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme</u>:

| | | | |
|---|---|---|---|
| *Full Marks* | : | $+4$ | **ONLY** if (all) the correct option(s) is(are) chosen; |
| *Partial Marks* | : | $+3$ | If all the four options are correct but **ONLY** three options are chosen; |
| *Partial Marks* | : | $+2$ | If three or more options are correct but **ONLY** two options are chosen, both of which are correct; |
| *Partial Marks* | : | $+1$ | If two or more options are correct but **ONLY** one option is chosen and it is a correct option; |
| *Zero Marks* | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| *Negative Marks* | : | $-2$ | In all other cases. |

Q.9 Consider the equation

$$\int_1^e \frac{(\log_e x)^{1/2}}{x\left(a-(\log_e x)^{3/2}\right)^2}dx = 1, \quad a \in (-\infty, 0) \cup (1, \infty).$$

Which of the following statements is/are TRUE ?

(A) **No** $a$ satisfies the above equation

(B) An integer $a$ satisfies the above equation

(C) An irrational number $a$ satisfies the above equation

(D) More than one $a$ satisfy the above equation

Q.10 Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression with $a_1 = 7$ and common difference 8. Let $T_1, T_2, T_3, \ldots$ be such that $T_1 = 3$ and $T_{n+1} - T_n = a_n$ for $n \geq 1$. Then, which of the following is/are TRUE ?

(A) $T_{20} = 1604$

(B) $\sum_{k=1}^{20} T_k = 10510$

(C) $T_{30} = 3454$

(D) $\sum_{k=1}^{30} T_k = 35610$

Q.11 Let $P_1$ and $P_2$ be two planes given by

$$P_1: \; 10x + 15y + 12z - 60 = 0\,,$$
$$P_2: \; -2x + 5y + 4z - 20 = 0\,.$$

Which of the following straight lines can be an edge of some tetrahedron whose two faces lie on $P_1$ and $P_2$ ?

(A) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$

(B) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(C) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$

(D) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$

Q.12 Let $S$ be the reflection of a point $Q$ with respect to the plane given by

$$\vec{r} = -(t+p)\hat{\imath} + t\hat{\jmath} + (1+p)\hat{k}$$

where $t, p$ are real parameters and $\hat{\imath}$, $\hat{\jmath}$, $\hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of $Q$ and $S$ are $10\hat{\imath} + 15\hat{\jmath} + 20\hat{k}$ and $\alpha\hat{\imath} + \beta\hat{\jmath} + \gamma\hat{k}$ respectively, then which of the following is/are TRUE ?

(A) $3(\alpha + \beta) = -101$

(B) $3(\beta + \gamma) = -71$

(C) $3(\gamma + \alpha) = -86$

(D) $3(\alpha + \beta + \gamma) = -121$

Q.13 Consider the parabola $y^2 = 4x$. Let $S$ be the focus of the parabola. A pair of tangents drawn to the parabola from the point $P = (-2,\ 1)$ meet the parabola at $P_1$ and $P_2$. Let $Q_1$ and $Q_2$ be points on the lines $SP_1$ and $SP_2$ respectively such that $PQ_1$ is perpendicular to $SP_1$ and $PQ_2$ is perpendicular to $SP_2$. Then, which of the following is/are TRUE ?

(A) $SQ_1 = 2$

(B) $Q_1Q_2 = \frac{3\sqrt{10}}{5}$

(C) $PQ_1 = 3$

(D) $SQ_2 = 1$

Q.14 Let $|M|$ denote the determinant of a square matrix $M$. Let $g: \left[0,\ \frac{\pi}{2}\right] \to \mathbb{R}$ be the function defined by

$$g(\theta) = \sqrt{f(\theta) - 1} + \sqrt{f\left(\frac{\pi}{2} - \theta\right) - 1}$$

where

$$f(\theta) = \frac{1}{2}\begin{vmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{vmatrix} + \begin{vmatrix} \sin\pi & \cos\left(\theta + \frac{\pi}{4}\right) & \tan\left(\theta - \frac{\pi}{4}\right) \\ \sin\left(\theta - \frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right) \\ \cot\left(\theta + \frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi \end{vmatrix}.$$

Let $p(x)$ be a quadratic polynomial whose roots are the maximum and minimum values of the function $g(\theta)$, and $p(2) = 2 - \sqrt{2}$. Then, which of the following is/are TRUE ?

(A) $p\left(\frac{3+\sqrt{2}}{4}\right) < 0$

(B) $p\left(\frac{1+3\sqrt{2}}{4}\right) > 0$

(C) $p\left(\frac{5\sqrt{2}-1}{4}\right) > 0$

(D) $p\left(\frac{5-\sqrt{2}}{4}\right) < 0$

**SECTION 3 (Maximum Marks: 12)**

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:
  *Full Marks* : $+3$ **ONLY** if the option corresponding to the correct combination is chosen;
  *Zero Marks* : $0$ If none of the options is chosen (i.e. the question is unanswered);
  *Negative Marks :* $-1$ In all other cases.

Q.15 Consider the following lists:

| **List-I** | **List-II** |
|---|---|
| (I) $\left\{x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3}\right] : \cos x + \sin x = 1\right\}$ | (P) has two elements |
| (II) $\left\{x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18}\right] : \sqrt{3}\tan 3x = 1\right\}$ | (Q) has three elements |
| (III) $\left\{x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5}\right] : 2\cos(2x) = \sqrt{3}\right\}$ | (R) has four elements |
| (IV) $\left\{x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4}\right] : \sin x - \cos x = 1\right\}$ | (S) has five elements |
| | (T) has six elements |

The correct option is:

(A) (I) → (P); (II) → (S); (III) → (P); (IV) → (S)

(B) (I) → (P); (II) → (P); (III) → (T); (IV) → (R)

(C) (I) → (Q); (II) → (P); (III) → (T); (IV) → (S)

(D) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)

Q.16 Two players, $P_1$ and $P_2$, play a game against each other. In every round of the game, each player rolls a fair die once, where the six faces of the die have six distinct numbers. Let $x$ and $y$ denote the readings on the die rolled by $P_1$ and $P_2$ , respectively. If $x > y$, then $P_1$ scores 5 points and $P_2$ scores 0 point. If $x = y$, then each player scores 2 points. If $x < y$, then $P_1$ scores 0 point and $P_2$ scores 5 points. Let $X_i$ and $Y_i$ be the total scores of $P_1$ and $P_2$, respectively, after playing the $i^{th}$ round.

| **List-I** | **List-II** |
|---|---|
| (I) Probability of $(X_2 \geq Y_2)$ is | (P) $\frac{3}{8}$ |
| (II) Probability of $(X_2 > Y_2)$ is | (Q) $\frac{11}{16}$ |
| (III) Probability of $(X_3 = Y_3)$ is | (R) $\frac{5}{16}$ |
| (IV) Probability of $(X_3 > Y_3)$ is | (S) $\frac{355}{864}$ |
| | (T) $\frac{77}{432}$ |

The correct option is:

(A) (I) → (Q); (II) → (R); (III) → (T); (IV) → (S)

(B) (I) → (Q); (II) → (R); (III) → (T); (IV) → (T)

(C) (I) → (P); (II) → (R); (III) → (Q); (IV) → (S)

(D) (I) → (P); (II) → (R); (III) → (Q); (IV) → (T)

Q.17 Let $p, q, r$ be nonzero real numbers that are, respectively, the $10^{th}$, $100^{th}$ and $1000^{th}$ terms of a harmonic progression. Consider the system of linear equations

$$\begin{aligned} x + y + z &= 1 \\ 10x + 100y + 1000z &= 0 \\ qr\,x + pr\,y + pq\,z &= 0 \quad . \end{aligned}$$

| List-I | List-II |
|---|---|
| (I) If $\frac{q}{r} = 10$, then the system of linear equations has | (P) $x = 0$, $y = \frac{10}{9}$, $z = -\frac{1}{9}$ as a solution |
| (II) If $\frac{p}{r} \neq 100$, then the system of linear equations has | (Q) $x = \frac{10}{9}$, $y = -\frac{1}{9}$, $z = 0$ as a solution |
| (III) If $\frac{p}{q} \neq 10$, then the system of linear equations has | (R) infinitely many solutions |
| (IV) If $\frac{p}{q} = 10$, then the system of linear equations has | (S) no solution |
| | (T) at least one solution |

The correct option is:

(A) (I) → (T); (II) → (R); (III) → (S); (IV) → (T)

(B) (I) → (Q); (II) → (S); (III) → (S); (IV) → (R)

(C) (I) → (Q); (II) → (R); (III) → (P); (IV) → (R)

(D) (I) → (T); (II) → (S); (III) → (P); (IV) → (T)

Q.18 Consider the ellipse

$$\frac{x^2}{4}+\frac{y^2}{3}=1\,.$$

Let $H(\alpha,0),\ 0<\alpha<2$, be a point. A straight line drawn through $H$ parallel to the $y$-axis crosses the ellipse and its auxiliary circle at points $E$ and $F$ respectively, in the first quadrant. The tangent to the ellipse at the point $E$ intersects the positive $x$-axis at a point $G$. Suppose the straight line joining $F$ and the origin makes an angle $\phi$ with the positive $x$-axis.

| **List-I** | **List-II** |
|---|---|
| (I) If $\phi=\frac{\pi}{4}$, then the area of the triangle $FGH$ is | (P) $\frac{\left(\sqrt{3}-1\right)^4}{8}$ |
| (II) If $\phi=\frac{\pi}{3}$, then the area of the triangle $FGH$ is | (Q) 1 |
| (III) If $\phi=\frac{\pi}{6}$, then the area of the triangle $FGH$ is | (R) $\frac{3}{4}$ |
| (IV) If $\phi=\frac{\pi}{12}$, then the area of the triangle $FGH$ is | (S) $\frac{1}{2\sqrt{3}}$ |
| | (T) $\frac{3\sqrt{3}}{2}$ |

The correct option is:

(A) (I) → (R); (II) → (S); (III) → (Q); (IV) → (P)

(B) (I) → (R); (II) → (T); (III) → (S); (IV) → (P)

(C) (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)

(D) (I) → (Q); (II) → (S); (III) → (Q); (IV) → (P)

## END OF THE QUESTION PAPER